

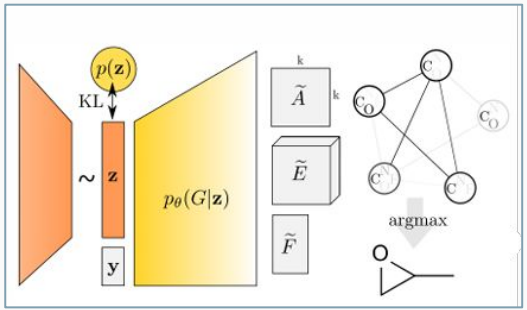
UNIVERSITÀ DI PISA

Generative and Deep Learning Project

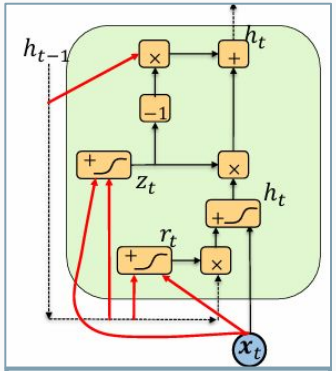
Curriculum: Artificial Intelligence
Student: Giorgio Chelli, 655949

Track

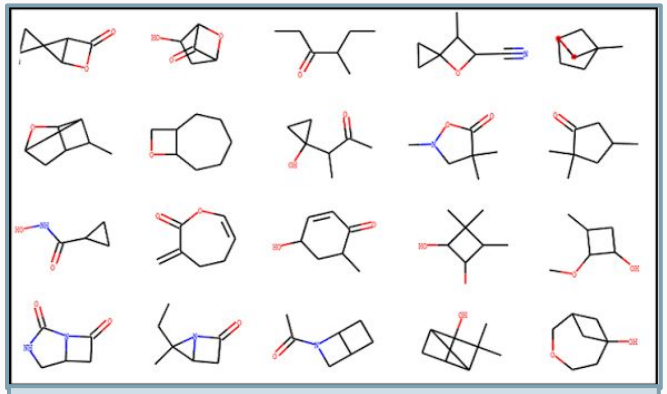
Track 9: Implement two different generative models for graph data of your choice (e.g. autoregressive, diffusion-based, VAE, …) and validate them on QM9 chemical graphs



GraphVAE



GRU



Q9 dataset examples

Selected Models:

- **GraphVAE**: *one-shot generation* of molecules
- **GRU**: *incremental generation* of molecules, using SMILES fragmentation or SPE

GraphVAE

➔ **Preprocessing settings:** As the QM9 dataset contains molecules with a variable number of heavy atoms (up to a maximum of 9), it was necessary to implement a zero-padding strategy to bring each structure back to a common dense representation

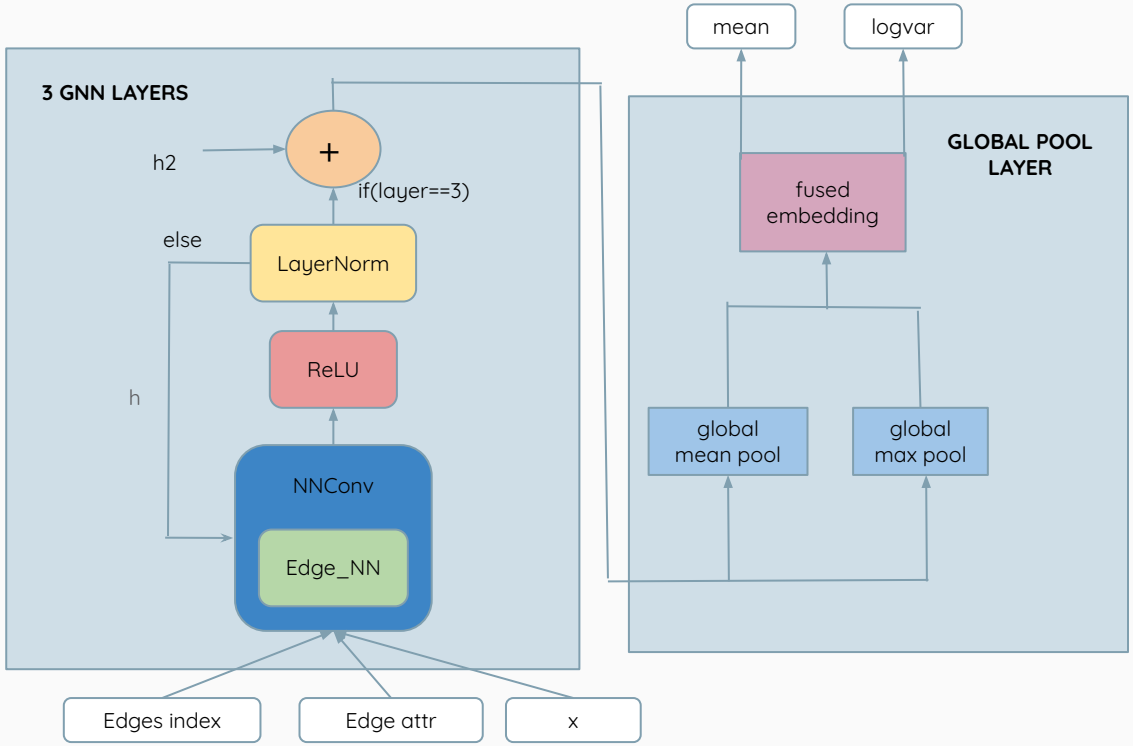
Nodes representation: A fixed-size vector of 9 elements (max number of heavy atoms)

Edges representation: A symmetric adjacency matrix (9x9)

➔ Since graphs are intrinsically invariant to node permutations, enforcing a rigid node ordering can introduce arbitrary biases during generation. To address this permutation ambiguity, during training we adopt a graph matching scheme based on the Hungarian algorithm

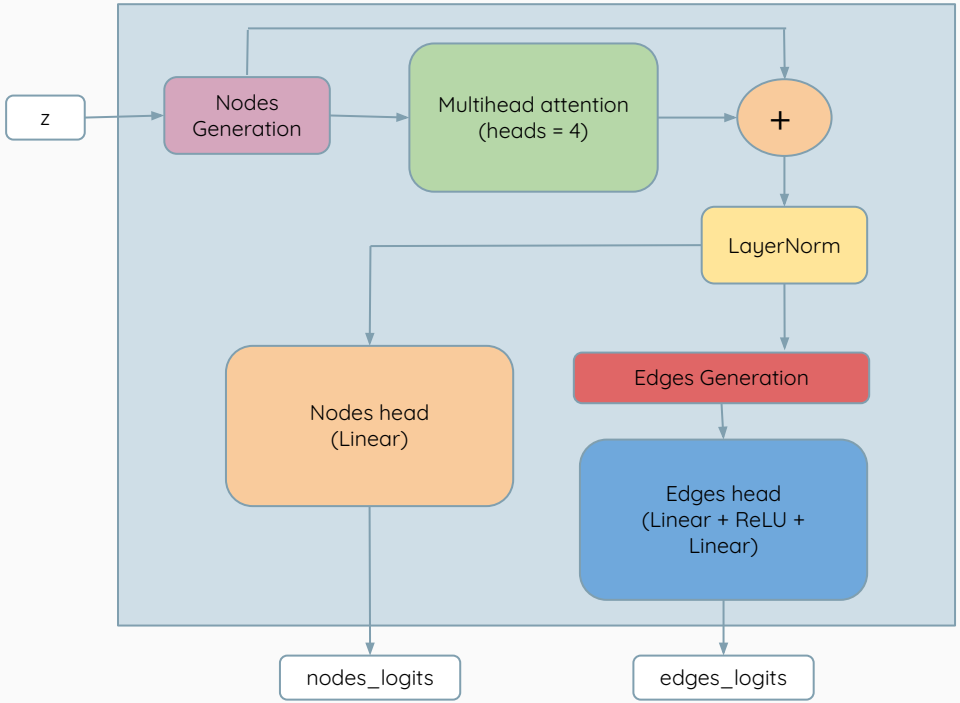
Encoder structure

- Dim_latent = 64
- Node_features = 5
- Edge_features = 4



Decoder structure

- Dim_latent = 64
- Dim_embedding = 256
- Node_features = 5
- Edge_features = 5
- Node_dim = 64
- Max_nodes = 9
- Heads = 4



Loss computation

- To train the GraphVAE model, the loss function is defined as a weighted linear combination of three components:

$$L_{total} = L_{nodes} + 2.0 \cdot L_{edges} + L_{valency}$$

- **Nodes loss**: The classification of the atoms associated with each node is optimised using a cross-entropy function.
- **Edges loss**: The topology of the links is evaluated by calculating the cross-entropy for each pair of nodes (i, j).
- **Valence loss**: This component introduces regularisation based on real chemical constraints, which is essential for the decoder to generate chemically stable and valid molecules.

Hungarian algorithm

- To calculate the reconstruction loss in graph generative models, one must account for graph isomorphism and permutation invariance. Following the GraphVAE framework, we align the predicted node and edge topologies with the target ground-truth graphs using the Hungarian Algorithm.
- The optimal permutation matrix P* for a graph with K potential nodes is computed by minimizing a bi-criterion assignment cost tensor $C \in R^{K \times K}$:

$$C_{i,j} = \alpha \cdot C_{i,j}^{\text{nodes}} + \beta \cdot C_{i,j}^{\text{edges}}$$

- where $C_{i,j}^{\text{nodes}}$ measures the cross-entropy loss between the predicted features of i-th node and j-th ground-truth node:

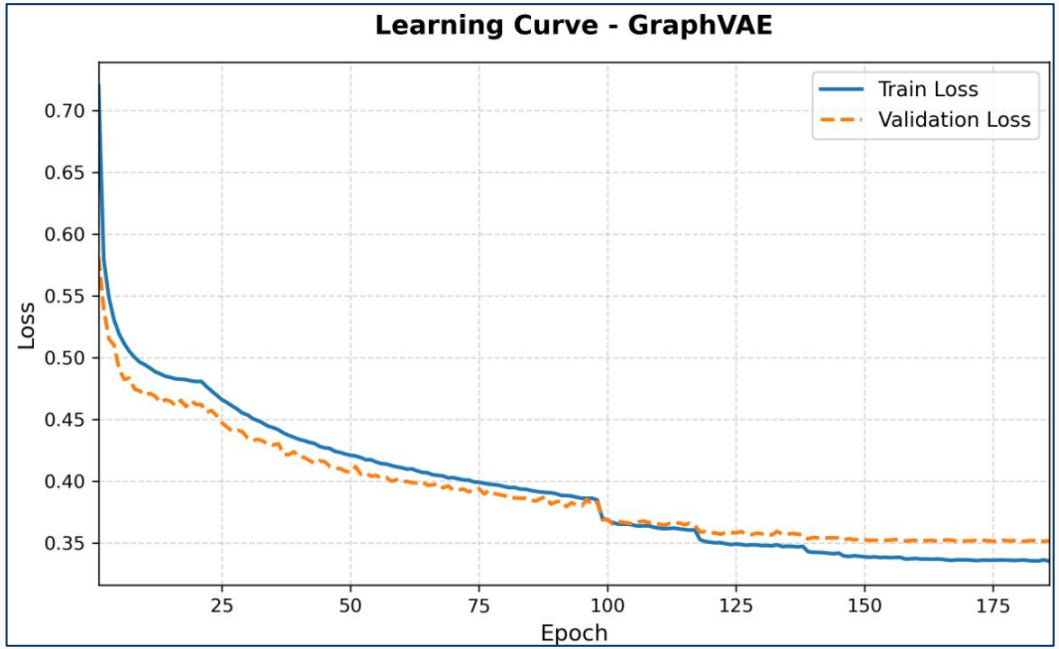
$$C_{i,j}^{\text{nodes}} = -\log P(\hat{X}_i = X_j^{\text{target}})$$

- To enforce stable convergence and preserve atomic semantics, we adopt $\beta = 0.0$ during training.

Training

- The GraphVAE model was trained for up to 180 epochs on the QM9 dataset, which was split into a training set (80%) and a validation set (20%), with the batch size set to 32.

  The Adam algorithm was used for optimisation, with an initial learning rate of 0.0005.



Generation

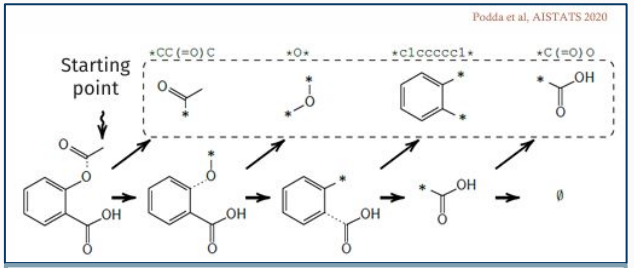
- New molecular structures are generated by sampling directly from the latent space and converting the resulting graphs into SMILES strings using the RDKit library. The decoder processes these vectors, returning the logits associated with nodes and edges. The logits are converted into probabilities using a Softmax function and sampled from a multinomial distribution.
- To limit the tendency of GraphVAE to create molecules as multiple disconnected fragments connectivity is enforced during testing by building a **maximum spanning tree (MST)** over the set of probable nodes, using the edge probabilities as similarity weights

Number of samples generated = 1000

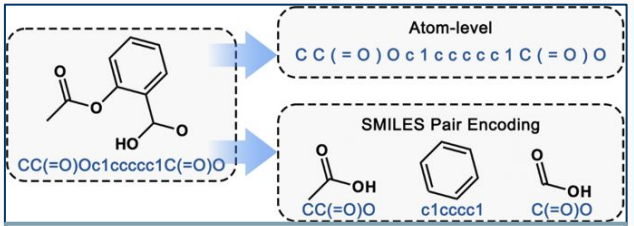
Temp.	Validity (%)	Uniqueness (%)	Novelty (%)
0.5	51.14 ± 1.99	99.36 ± 0.44	93.78 ± 0.18
1.0	51.52 ± 1.63	99.65 ± 0.23	94.83 ± 0.99
1.5	50.72 ± 2.39	99.73 ± 0.19	94.46 ± 0.26

GRU

- **Preprocessing settings:** To process the molecules using the GRU, the structures were initially converted into SMILES strings and subsequently processed using the fragmentation algorithm.
  Two different and contrasting approaches for fragmentation were used and compared:
  - **Fragmentation Algorithm based on BRICS**: Directly applying BRICS fragmentation to the QM9 dataset revealed significant structural limitations. BRICS relies on a predefined set of cleavage rules. QM9 consists exclusively of small, highly constrained molecules; most structures lack the target functional groups required by BRICS rules.
  - **SMILES Pair Encoding (SPE)**: SPE adapts the Byte Pair Encoding (BPE) algorithm specifically for SMILES representations, iteratively merging the most frequent pairs of adjacent characters or subsequences into unified tokens.



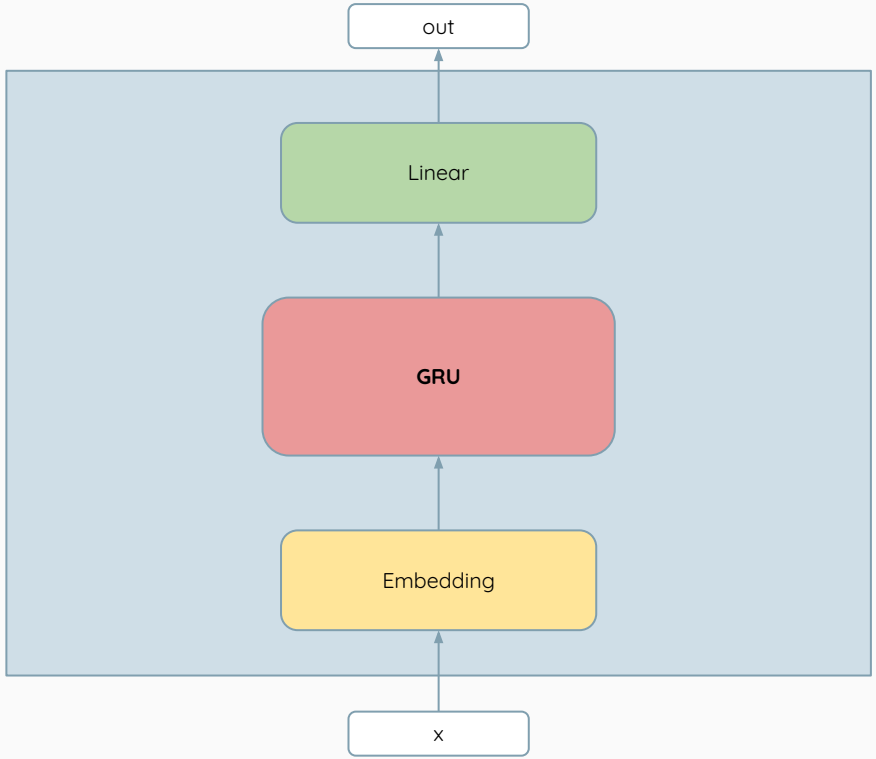
BRICS fragmentation



SPE fragmentation

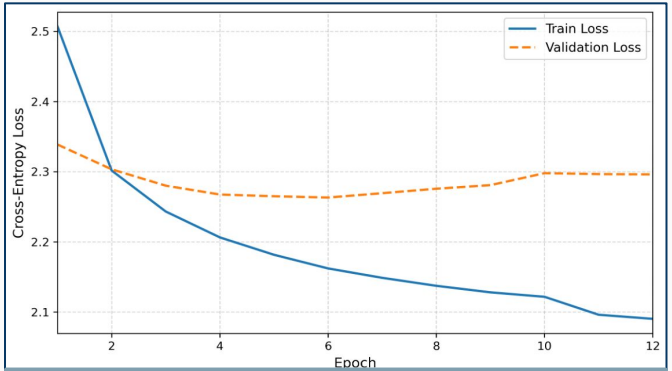
GRU structure

- Dim_hidden = 512
- Dim_embedding = 256
- Num_layers = 3
- Dropout = 0.2

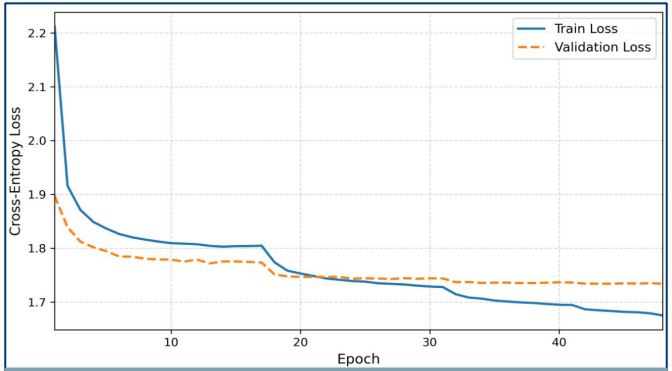


Training

The GraphGRU model consists of three recurrent layers with an embedding dimension of 256. The dataset was split into a training set (80%) and a validation set (20%), with the batch size set to 32. Cross-entropy loss was used to guide the autoregressive learning process; this explicitly excludes padding tokens from the gradient calculation.
The optimisation process utilises the Adam algorithm with an initial learning rate of 0.0001.



Training curve with BRICS fragmentation



Training curve with SPE fragmentation

Generation

➔ The autoregressive generation pipeline evaluates candidate models by sampling token sequences from the learned conditional probability distribution using temperature-scaled multinomial sampling. However, reconstructing a valid molecular graph from the generated token sequence reveals key operational differences between the BRICS and SPE tokenization schemes.

Number of samples generated = 1000

Fragmentation	Temp.	Validity (%)	Uniqueness (%)	Novelty (%)
BRICS	0.5	60.52 ± 0.80	98.71 ± 0.39	1.28 ± 0.42
	1.0	52.48 ± 0.35	88.07 ± 0.74	10.60 ± 2.09
	1.5	40.52 ± 0.82	85.49 ± 0.95	33.54 ± 2.29
SPE	0.5	83.92 ± 0.64	95.61 ± 0.41	3.69 ± 0.35
	1.0	75.62 ± 1.27	99.63 ± 0.15	17.09 ± 0.80
	1.5	61.04 ± 1.85	99.90 ± 0.13	33.05 ± 1.36

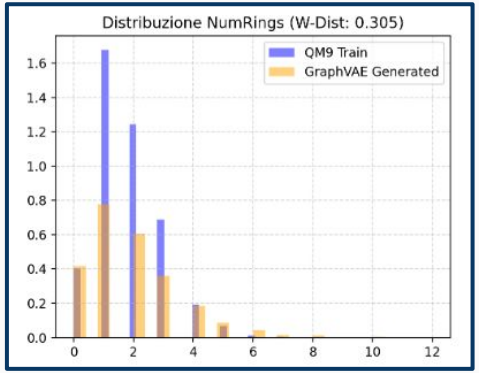
Wasserstein distance analysis

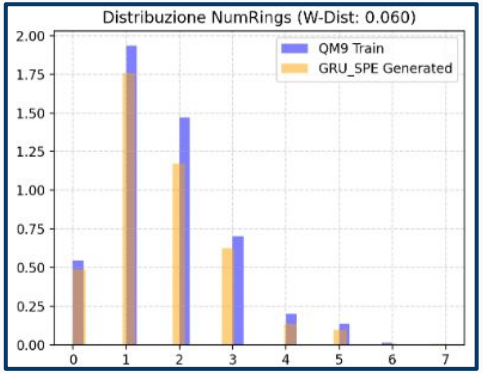
- GraphGRU (SPE) achieves the lowest Wasserstein distances for almost all descriptors, the closest alignment with QM9 observed in this study. This result is achieved despite a non-negligible novelty rate, indicating that the novelty of SPE reflects genuine exploration within a chemically realistic distribution.
- GraphGRU (BRICS) exhibits the opposite pattern: despite very low novelty at low temperatures, its Wasserstein distances remain relatively sparse. The model overwhelmingly reproduces fragments nearly identical to the BRICS vocabulary, while the fragments it recombines novelly tend to distort the overall molecular weight and polarity profile.
- In GraphVAE high novelty occurs in conjunction with the largest Wasserstein distance on the ring count and a substantially higher TPSA distance compared to GraphGRU-SPE. This distributionally confirms that GraphVAE's novelty is partly driven by structurally atypical generation rather than by the exploration of regions of molecular space chemically consistent with QM9.

Model	MW	LogP	TPSA	QED	NumRings
GraphVAE	2.3427	**0.1670**	3.2496	**0.0076**	0.3226
GraphGRU (BRICS)	11.7337	0.3966	4.7542	0.0140	0.2351
GraphGRU (SPE)	**0.4056**	0.2231	**1.3196**	0.0117	**0.0590**

Root Cause: Aromatic Bond Collapse

➔ Examining the high novelty of GraphVAE, we found that the model systematically underproduces aromatic bonds on heteroatoms. This shortcoming stems from the decoder's independent predictions for each individual edge: aromatic rings require simultaneous coherence across 5-6 edges to pass RDKit's kekulization check, which independent sampling for each edge rarely achieves, even when individual bond probabilities are well-calibrated. This limitation is not specific to our implementation: the original formulation of GraphVAE explicitly assumes independence among all predicted variables. This same mechanism also explains the anomalous ring count distribution, since ring count directly depends on multi-edge coherence.





Conclusions

In summary, **GraphGRU** with SPE fragmentation emerges as the most balanced and effective framework for valid and constrained molecular design, while **GraphVAE** serves as a highly capable exploratory baseline for continuous latent space optimization despite structurally suffers from the lack of explicit connectivity constraints in the decode. These limitations lead to the use of a different distribution in generation (primarily in terms of the number of rings) compared to QM9, exponentially increasing the novelty values.

Example of 10 molecules generated at a temperature of 1.5 using GraphGRU and SPE fragmentation.

